

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 30%

QUESTIONS: 05 Total Marks:50

Annexure A

MOCK INTERVIEW

<i>Criteria</i>	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	<i>Communication(1.5)</i>	Confidence (1)	Response to Question	<i>Total(10)</i>
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<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts

Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

(Each question aligned with BL3–BL4, 0.75 Marks each)

Q1. Dataset: Retail Transactions

TID Items

T1 Milk, Bread

T2 Milk, Butter

T3 Bread, Butter
T4 Milk, Bread, Butter
T5 Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example. (*BL3 – 0.75 Marks*)

Q2. Dataset: Transaction Data

TID Items

T1 A, B
T2 B, C
T3 A, C
T4 A, B, C
T5 B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used. (*BL4 – 0.75 Marks*)

Q3. Dataset: Transaction Data

TID Items

T1 A, B
T2 A
T3 B
T4 A, B
T5 A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining. (*BL3 – 0.75 Marks*)

Q4 . Dataset: Multilevel Data

TID Items

T1 Electronics, Mobile
T2 Electronics, Laptop
T3 Electronics, Mobile
T4 Electronics, Mobile
T5 Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional rules. (BL4 – 0.75 Marks)

Q5. Dataset: Large Transaction Sample

TID Items

T1 A, B, C

T2 A, B

T3 B, C

T4 A, C

T5 A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues. (BL4 – 0.75 Marks)

SIMATS ENGINEERING

CSA16 – Data Warehousing and Data Mining

Assessment Tool 3 – Annexure A: Mock Interview – Model Answers

Q1. Association Rule Mining to Improve Retail Sales

Question: Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example. (BL3 – 0.75 Marks)

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Association rule mining discovers items that are frequently purchased together by analyzing transaction (market basket) data. It generates rules of the form $X \rightarrow Y$, each measured by **support** (how often X and Y occur together) and **confidence** (how often Y is bought when X is bought). Retailers use these rules to understand hidden purchase patterns rather than guessing what customers buy together.

In the given dataset, Milk and Bread appear together in 2 of 5 transactions (support = 0.4), and whenever Milk is bought, Bread is bought 50% of the time (confidence = $0.4/0.8 = 50\%$). Similarly, Bread

and Butter occur together with a confidence of about 66.7% (0.4/0.6), suggesting Bread → Butter is a fairly strong rule.

Practical ways this improves sales:

- **Product placement:** Placing Bread and Butter near each other (or near Milk) increases the chance of a combined purchase, since the rule shows they co-occur often.
- **Cross-selling and bundling:** Offering a "Milk + Bread" combo discount encourages customers to buy both, increasing basket size.
- **Targeted promotions:** A customer who adds Milk to their cart (online store) can be shown a recommendation for Bread or Butter based on the discovered rule.
- **Inventory and shelf planning:** Frequently co-purchased items can be stocked together to reduce walking time and improve customer experience, indirectly boosting sales.

In short, association rule mining converts raw transaction logs into actionable cross-selling strategies, helping the retailer increase average order value and customer satisfaction.

Q2. Apriori vs FP-Growth

Question: Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used. (BL4 – 0.75 Marks)

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

For this dataset (5 transactions), $\text{support}(A) = 3/5 = 0.6$, $\text{support}(B) = 4/5 = 0.8$, $\text{support}(C) = 4/5 = 0.8$, $\text{support}(A,B) = 2/5 = 0.4$, $\text{support}(B,C) = 3/5 = 0.6$ and $\text{support}(A,C) = 2/5 = 0.4$. Both algorithms would arrive at the same frequent itemsets and rules, but they reach that result very differently.

Apriori uses a candidate generate-and-test, level-wise approach. It first finds frequent 1-itemsets (A, B, C), then joins them to form candidate 2-itemsets (AB, AC, BC), scans the database again to compute their support, prunes the infrequent ones using the Apriori property (a superset of an infrequent itemset cannot be frequent), and repeats for 3-itemsets. This requires multiple full scans of the transaction database (one scan per itemset size) and can generate a very large number of candidate itemsets when the dataset or the number of distinct items is large.

FP-Growth avoids candidate generation entirely. It scans the database twice: once to find item frequencies (ordering B and C ahead of A here, since they are more frequent), and once to build a compact FP-tree that stores the transactions in prefix-sharing paths (e.g., T2, T3, T4 and T5 share the

B-C / A-C branches). Frequent itemsets are then mined directly from this tree using a divide-and-conquer, pattern-growth strategy, without ever generating candidate itemsets explicitly.

When to use each:

- **Use Apriori** when the dataset is small to moderate in size, when simplicity and ease of implementation/interpretation matter, or when memory is limited (Apriori does not need to hold a tree structure in memory, only candidate lists).
- **Use FP-Growth** when the dataset is large or very dense (many items per transaction, or low minimum support), since repeated database scans in Apriori become a major bottleneck. FP-Growth is generally faster and more scalable for big data warehousing scenarios because it compresses the database into a tree and mines it in memory.

Q3. Support and Confidence

Question: Explain support and confidence using the dataset and describe their importance in association rule mining. (BL3 – 0.75 Marks)

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B

T5	A
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Support measures how frequently an itemset appears in the dataset, i.e. $\text{Support}(X) = (\text{Number of transactions containing } X) / (\text{Total number of transactions})$. It tells us how popular or common an itemset is across all transactions.

Confidence measures how often the rule $X \rightarrow Y$ is found to be true, i.e. $\text{Confidence}(X \rightarrow Y) = \text{Support}(X, Y) / \text{Support}(X)$. It tells us, given that a customer bought X, how likely they are to also buy Y.

Calculations for this dataset (5 transactions):

- $\text{Support}(A) = 4/5 = 0.8$ (A appears in T1, T2, T4, T5)
- $\text{Support}(B) = 3/5 = 0.6$ (B appears in T1, T3, T4)
- $\text{Support}(A, B) = 2/5 = 0.4$ (A and B occur together in T1 and T4)
- $\text{Confidence}(A \rightarrow B) = \text{Support}(A, B) / \text{Support}(A) = 0.4 / 0.8 = 0.5 = 50\%$
- $\text{Confidence}(B \rightarrow A) = \text{Support}(A, B) / \text{Support}(B) = 0.4 / 0.6 \approx 0.667 = 66.7\%$

This shows that although A and B occur together in the same number of transactions either way, the rule

$B \rightarrow A$ is stronger (66.7% confidence) than $A \rightarrow B$ (50% confidence), because B is less frequent overall so its co-occurrence with A is proportionally more significant.

Importance:

- Support filters out rare, statistically insignificant itemsets, so mining focuses only on patterns that occur often enough to be meaningful and actionable.
- Confidence measures the reliability/strength of a rule, ensuring that only itemsets with a strong predictive relationship are reported as rules.
- Together, minimum support and minimum confidence thresholds control the number and quality of rules generated, balancing completeness with relevance and avoiding information overload for decision-makers.

Q4. Multilevel vs Multidimensional Association Rules

Question: Explain multilevel association rules using the dataset and compare them with multidimensional rules. (BL4 – 0.75 Marks)

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Multilevel association rules are mined across a concept hierarchy, where items are organized from general (top) to specific (bottom) categories — for example Electronics (level 1) $\rightarrow$ Mobile / Laptop (level 2) $\rightarrow$ specific brand/model (level 3). In this dataset, every transaction contains "Electronics" (support = $5/5$ = 1.0), while "Mobile" appears in $3/5 = 0.6$ and "Laptop" in $2/5 = 0.4$ of transactions. A rule at the top level, such as Electronics $\rightarrow$ (frequent in all baskets), is too general to be useful, but drilling down reveals a more specific and useful pattern: customers who buy Electronics tend to specifically buy Mobile (60% of the time) more than Laptop (40%). This is the essence of multilevel mining — rules can be generated at different levels of abstraction, and lower/more specific levels often reveal patterns that a single flat-level analysis would hide, while very general top-level rules are filtered out for being trivial.

Multilevel mining typically uses a reduced minimum support at lower (more specific) levels, since specific items naturally occur less often than their general category.

Multidimensional association rules, in contrast, involve multiple independent attributes/dimensions (not a single hierarchy of one attribute) such as Item, Age Group, Location, and Time. A multidimensional rule might look like: "Customers aged 20–30 in Chennai who buy Electronics also buy Mobile phones on weekends." Here Age, Location and Time are separate dimensions combined with the Item dimension,

rather than levels of the same item hierarchy.

Key comparison:

- Multilevel rules operate **within one dimension** (item) across a concept hierarchy (general to specific); multidimensional rules operate **across several different dimensions/attributes** simultaneously.
- Multilevel mining is mainly concerned with the right level of abstraction and support thresholds per level; multidimensional mining is concerned with combining categorical and quantitative attributes together.
- In this dataset, a multilevel rule is "Electronics → Mobile"; a multidimensional version would add another attribute, e.g. "Electronics, Weekday → Mobile."

Q5. Challenges in Mining Large Datasets and Scalable Methods

Question: Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues. (BL4 – 0.75 Marks)

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

In this small sample, $\text{Support}(A) = \text{Support}(B) = \text{Support}(C) = 4/5 = 0.8$ and $\text{Support}(A,B,C) = 2/5 = 0.4$ (T1 and T5). With only 5 transactions and 3 items this is trivial to compute, but the same logic scaled to millions of transactions and thousands of items — as in a real retail data warehouse — exposes several challenges:

- **Combinatorial explosion:** The number of candidate itemsets grows exponentially with the number of distinct items, making level-wise candidate generation (as in Apriori) computationally very expensive for large item sets.
- **Multiple database scans:** Algorithms like Apriori require one full pass over the data for every itemset size, which is extremely costly when the transaction table has millions of rows.
- **Memory limitations:** Storing all candidate itemsets and their counts, or an entire FP-tree, may exceed available memory for very large or very dense datasets.
- **I/O and processing time:** Disk-based access to huge transaction logs (typical in a data warehouse) is slow, and repeated scanning multiplies this cost.
- **Data skew and distribution:** When data is stored/processed across distributed systems, uneven

data distribution across nodes can create load-balancing problems.

Scalable methods address these challenges as follows:

- **FP-Growth** compresses the database into an FP-tree in only two scans and mines frequent patterns without candidate generation, greatly reducing both I/O and computation compared to Apriori.
- **Partitioning-based algorithms** (e.g., Partition algorithm) split the database into smaller chunks that fit in memory, mine each partition locally, and then combine/verify the results globally, reducing the number of full database scans to two.
- **Sampling-based approaches** mine frequent itemsets from a representative random sample of the data rather than the entire dataset, trading a small amount of accuracy for a large gain in speed.
- **Parallel and distributed mining** (e.g., MapReduce/Spark-based Apriori or FP-Growth) distributes the workload across multiple nodes, allowing large-scale data warehouses to mine billions of transactions within practical time limits.
- **Vertical data format mining** (e.g., Eclat algorithm) stores data as item → transaction-ID lists and computes support via fast set intersections, avoiding repeated horizontal database scans.

Overall, scalable methods reduce the number of database scans, avoid explicit candidate generation where possible, and exploit compression, partitioning, sampling, or distributed computing so that association rule mining remains feasible as data volume and dimensionality grow.